MakeMyMarrige

DATABASE SCHEMA DESIGN & DIAGRAMS

Version 1.0 | November 2026

MongoDB Atlas • Hybrid Modelling • 82 Collection Specifications • 5 Diagrams

1. Architecture baseline

• Shared database/shared collections; logical tenant isolation through weddingId.

• One collection per independently managed entity; bounded parent-owned values may be embedded.

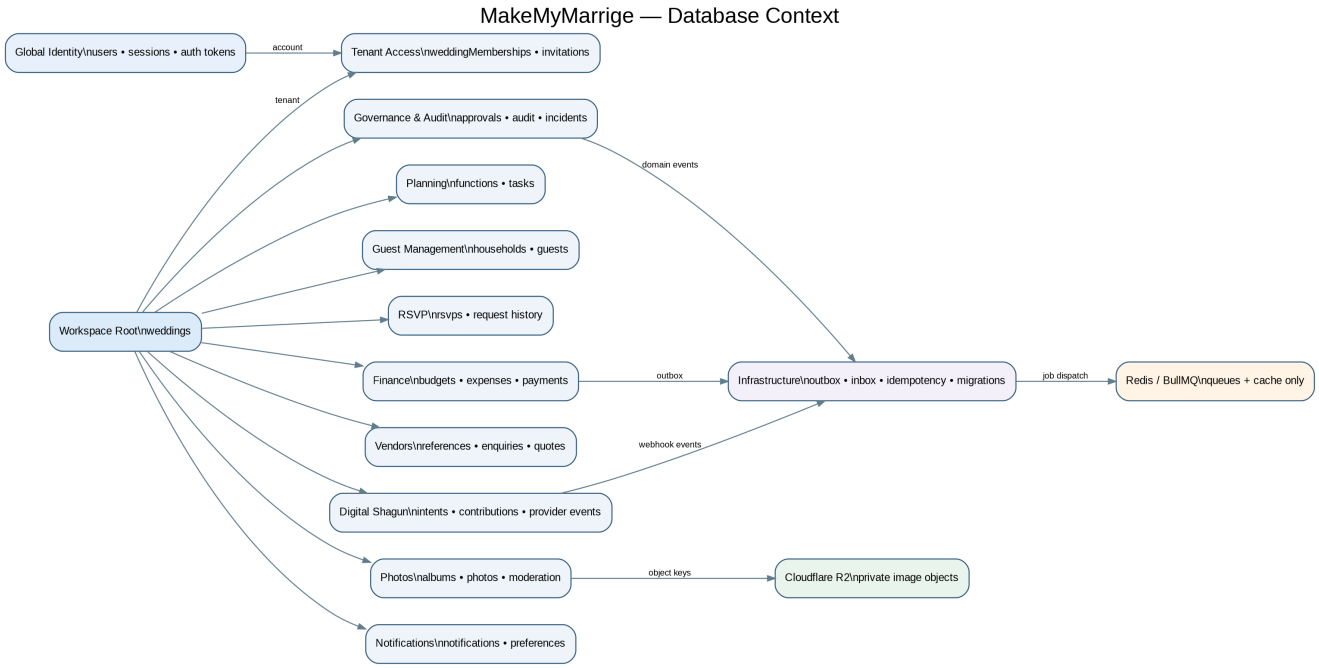
• Global identity/session data is separate from tenant-owned wedding data.

• ObjectId internally; opaque hashed tokens externally. Money in integer paise; date-only YYYY-MM-DD; instants UTC BSON Date.

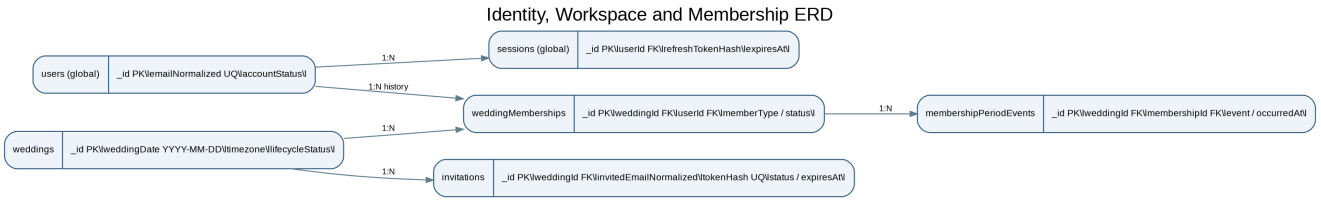
• Three-layer validation, OCC, transactions, transactional outbox, idempotency and selective TTL.

2. Database context and ER diagrams

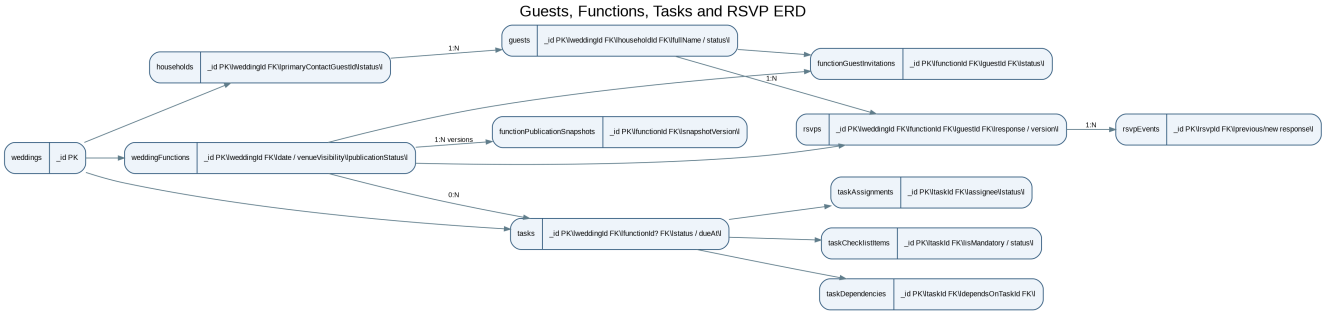
System context and domain boundaries



Global identity, workspace root, membership and invitation relationships



Households, guests, functions, task and RSVP relationships



Budget, expense, vendor and Digital Shagun relationships

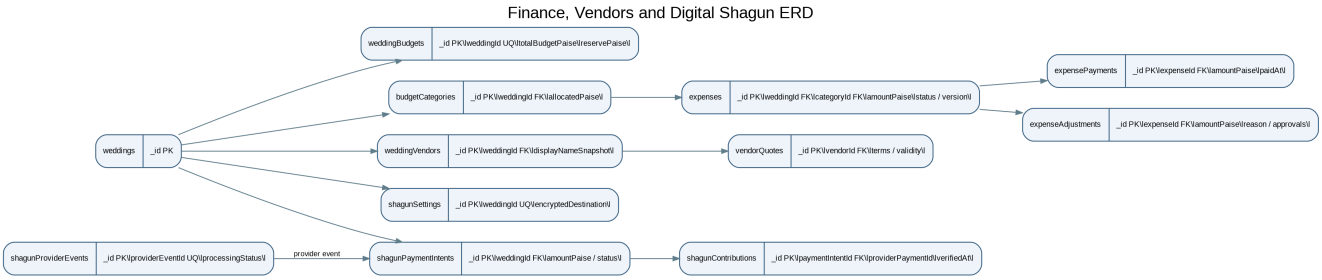
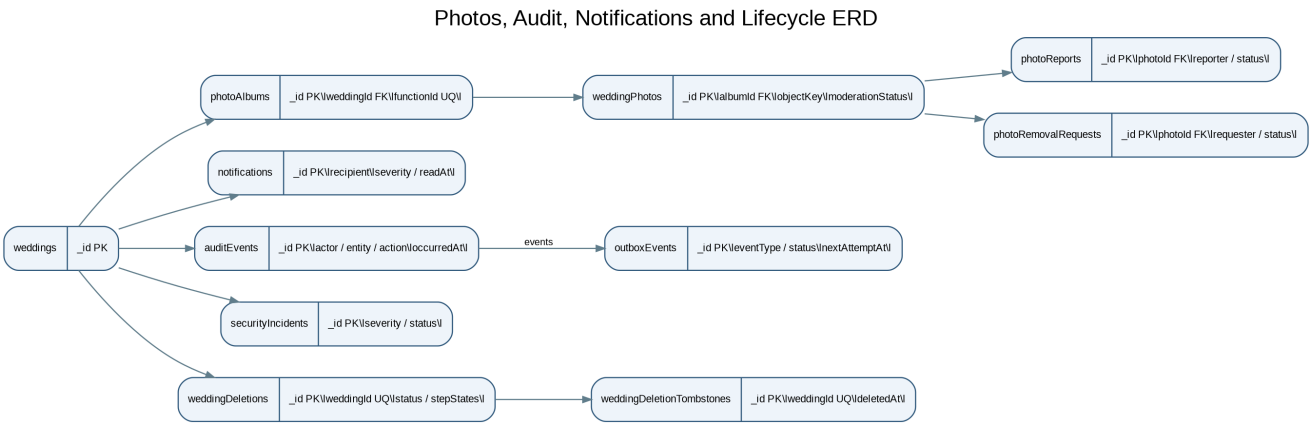


Photo moderation, notifications, audit and deletion relationships



3. Collection inventory

Identity / global: users, sessions, emailVerificationTokens, passwordResetTokens

Workspace: weddings, weddingMemberships, membershipPeriodEvents, invitations

Governance: governanceRequests, weddingFinanceSettings, permissionChangeEvents

Planning: weddingFunctions, functionPublicationSnapshots, functionHouseholdInvitations, functionGuestInvitations, tasks, taskAssignments, taskChecklistItems, taskDependencies, taskComments, taskActivityEvents

Guests: households, guests, guestRelationshipTags, guestCorrectionRequests, guestMergeOperations, guestHouseholdHistory

RSVP: rsvps, rsvpEvents, additionalGuestRequests, additionalGuestRequestEvents

Finance: weddingBudgets, budgetCategories, budgetGovernanceRequests, budgetReserveAllocations, expenseEstimates, vendorQuotes, expenses, expenseApprovalEvents, expensePayments, expenseAdjustments, financeAuditEvents

Vendors: vendorReferences, weddingVendors, vendorCategories, vendorServices, vendorEnquiries, vendorEstimates, vendorAvailabilityChecks, vendorSelections, vendorBookingRecords, vendorActivityEvents

Shagun: shagunSettings, shagunConfigurationRequests, shagunPaymentIntents, shagunContributions, shagunProviderEvents, shagunReconciliationEvents, shagunAcknowledgements, shagunAuditEvents

Photos: photoAlbums, weddingPhotos, photoReports, photoRemovalRequests, photoAuditEvents

Notifications: notifications, notificationPreferences, dashboardSnapshots (optional)

Audit/lifecycle: auditEvents, securityIncidents, dataAccessRequests, dataCorrectionRequests, weddingLifecycleEvents, weddingDeletions, weddingDeletionTombstones, weddingExportRequests, cancellationNoticeRecords

Platform: supportRecoveryCases

Infrastructure: outboxEvents, inboxEvents, idempotencyRecords, migrationHistory

4. Detailed collection schema catalogue

users — Identity / global

Fields: emailNormalized:string unique; passwordHash:string?; emailVerifiedAt:Date?; displayName:string; accountStatus:ACTIVE|SUSPENDED|CLOSED; authProviderLinks:bounded array; createdAt; updatedAt; version:int.

Indexes / constraints: Unique emailNormalized. Never embed sessions.

sessions — Identity / global

Fields: userId:ObjectId; refreshTokenHash:string; sessionFamilyId:string; expiresAt:Date; revokedAt:Date?; lastUsedAt:Date?; deviceLabel:string?; minimal IP metadata; createdAt.

Indexes / constraints: Index userId+revokedAt+expiresAt. TTL only after expiry policy.

emailVerificationTokens — Identity / global

Fields: userId:ObjectId; tokenHash:string; expiresAt:Date; consumedAt:Date?; createdAt:Date.

Indexes / constraints: Unique tokenHash; TTL expiresAt; enforce expiry in API.

passwordResetTokens — Identity / global

Fields: userId:ObjectId; tokenHash:string; expiresAt:Date; consumedAt:Date?; createdAt:Date.

Indexes / constraints: Unique tokenHash; TTL expiresAt; single-use.

weddings — Workspace

Fields: name:string; weddingDate:YYYY-MM-DD; timezone:'Asia/Kolkata'; lifecycleStatus enum; deletionDueAt:Date?; createdAt; updatedAt; version:int.

Indexes / constraints: Workspace root; no embedded business collections.

weddingMemberships — Workspace

Fields: weddingId:ObjectId; userId:ObjectId; memberType:CO_OWNER|ORGANISER|TREASURER; status:ACTIVE|SUSPENDED|ENDED; permissions:bounded grants; joinedAt; endedAt?; version:int.

Indexes / constraints: Partial unique userId for ACTIVE/SUSPENDED; weddingId+status+memberType.

membershipPeriodEvents — Workspace

Fields: weddingId:ObjectId; membershipId:ObjectId; eventType:JOINED|SUSPENDED|REACTIVATED|ENDED; actorId; reason?; occurredAt.

Indexes / constraints: Append-only; weddingId+membershipId+occurredAt.

invitations — Workspace

Fields: weddingId:ObjectId; invitedEmailNormalized:string; invitationType; tokenHash; status:PENDING|ACCEPTED|REVOKED|EXPIRED; expiresAt; createdBy; acceptedBy?; acceptedAt?; createdAt.

Indexes / constraints: Unique tokenHash; weddingId+email+status; retain history.

governanceRequests — Governance

Fields: weddingId:ObjectId; requestType; targetEntityType; targetEntityId:ObjectId; proposedSnapshot:bounded object; proposalVersion:int; approvals:bounded distinct co-owner approvals; status; expiresAt; createdBy; createdAt; version:int.

Indexes / constraints: Index weddingId+status+createdAt; proposal changes invalidate approvals.

weddingFinanceSettings — Governance

Fields: weddingId:ObjectId unique; treasurerMembershipId:ObjectId?; approvalPolicyVersion:int; reserveApprovalThresholdPaise:int64?; updatedAt; version:int.

Indexes / constraints: One settings record per wedding; changes follow dual-owner governance.

permissionChangeEvents — Governance

Fields: weddingId:ObjectId; membershipId:ObjectId; changedBy; before/after permission snapshot; reason; occurredAt.

Indexes / constraints: Append-only audit of permission changes.

weddingFunctions — Planning

Fields: weddingId:ObjectId; name; date:YYYY-MM-DD; startTime/endTime:HH:mm?; venueVisibility:ANNOUNCED|TO_BE_ANNOUNCED|PRIVATE; draftDetails:bounded; publicationStatus:DRAFT|PUBLISHED|CANCELLED; version:int; timestamps.

Indexes / constraints: weddingId+date; status/date queries.

functionPublicationSnapshots — Planning

Fields: weddingId:ObjectId; functionId:ObjectId; snapshotVersion:int; publishedSnapshot:immutable bounded object; approvedBy:[two distinct co-owner IDs]; publishedAt:Date.

Indexes / constraints: Unique weddingId+functionId+snapshotVersion.

functionHouseholdInvitations — Planning

Fields: weddingId:ObjectId; functionId:ObjectId; householdId:ObjectId; status:INVITED|REVOKED; invitedAt; createdBy.

Indexes / constraints: Unique weddingId+functionId+householdId.

functionGuestInvitations — Planning

Fields: weddingId:ObjectId; functionId:ObjectId; guestId:ObjectId; householdInvitationId:ObjectId?; status:INVITED|REVOKED; invitedAt; createdBy.

Indexes / constraints: Unique weddingId+functionId+guestId.

tasks — Planning

Fields: weddingId:ObjectId; functionId:ObjectId?; title; description?; status:TODO|IN_PROGRESS|BLOCKED|COMPLETED|CANCELLED; priority; dueAt:Date?; createdBy; version:int; timestamps.

Indexes / constraints: weddingId+status+dueAt+_id; weddingId+functionId+status.

taskAssignments — Planning

Fields: weddingId:ObjectId; taskId:ObjectId; assigneeMembershipId:ObjectId; assignedBy; assignedAt; status:ACTIVE|REMOVED.

Indexes / constraints: Partial unique weddingId+taskId+assigneeMembershipId for ACTIVE.

taskChecklistItems — Planning

Fields: weddingId:ObjectId; taskId:ObjectId; title; isMandatory:boolean; status:OPEN|COMPLETED; completedBy:ObjectId?; completedAt:Date?; sortOrder:int; version:int.

Indexes / constraints: weddingId+taskId+sortOrder. Separate entity for independently validated completion.

taskDependencies — Planning

Fields: weddingId:ObjectId; taskId:ObjectId; dependsOnTaskId:ObjectId; createdBy; createdAt.

Indexes / constraints: Unique weddingId+taskId+dependsOnTaskId; reject self-edge and cycles in domain service.

taskComments — Planning

Fields: weddingId:ObjectId; taskId:ObjectId; authorId:ObjectId; body:string; createdAt; editedAt?; editReason?

Indexes / constraints: weddingId+taskId+createdAt; comments append-oriented.

taskActivityEvents — Planning

Fields: weddingId:ObjectId; taskId:ObjectId; actorId; action; beforeSummary?; afterSummary?; occurredAt.

Indexes / constraints: Append-only; weddingId+taskId+occurredAt.

households — Guests

Fields: weddingId:ObjectId; label:string; primaryContactGuestId:ObjectId?; status:ACTIVE|ARCHIVED; createdAt; updatedAt; version:int.

Indexes / constraints: weddingId+status; primary contact must be active guest in same household.

guests — Guests

Fields: weddingId:ObjectId; householdId:ObjectId?; fullName:string; phoneNormalized:string?; phoneLookupHmac:string?; emailNormalized:string?; status:ACTIVE|ARCHIVED; identityVerificationState; primaryRelationshipTagId?; secondaryRelationshipTagIds:bounded array; timestamps; version:int.

Indexes / constraints: weddingId+householdId+status; avoid global phone uniqueness because shared numbers allowed.

guestRelationshipTags — Guests

Fields: weddingId:ObjectId; name:string; tagType:PRIMARY|SECONDARY; isActive:boolean; createdAt; updatedAt.

Indexes / constraints: Unique normalised name per wedding/type if required.

guestCorrectionRequests — Guests

Fields: weddingId:ObjectId; guestId:ObjectId; requestedChanges:bounded object; requestedBy; status:PENDING|APPROVED|REJECTED|CANCELLED; reviewedBy?; reason?; timestamps.

Indexes / constraints: Requester cannot approve own request.

guestMergeOperations — Guests

Fields: weddingId:ObjectId; sourceGuestId:ObjectId; targetGuestId:ObjectId; requestedBy; approvals:[two co-owners]; status; createdAt; completedAt?; migrationSummary?

Indexes / constraints: Append-only operation record; preserve source guest history.

guestHouseholdHistory — Guests

Fields: weddingId:ObjectId; guestId:ObjectId; fromHouseholdId:ObjectId?; toHouseholdId:ObjectId?; changedBy; reason; changedAt.

Indexes / constraints: Append-only; weddingId+guestId+changedAt.

rsvps — RSVP

Fields: weddingId:ObjectId; functionId:ObjectId; guestId:ObjectId; response:NOT_YET_RESPONDED|ATTENDING|NOT_ATTENDING; respondedAt:Date?; source:GUEST|ORGANISED_ASSISTANCE; version:int; timestamps.

Indexes / constraints: Unique weddingId+functionId+guestId.

rsvpEvents — RSVP

Fields: weddingId:ObjectId; rsvpId:ObjectId; functionId:ObjectId; guestId:ObjectId; previousResponse?; newResponse; actorId; actorType; reason?; occurredAt.

Indexes / constraints: Append-only; weddingId+functionId+occurredAt.

additionalGuestRequests — RSVP

Fields: weddingId:ObjectId; functionId:ObjectId; requestingGuestId:ObjectId; requestedCount:int; approvedCount:int; status:PENDING_ARRANGEMENT|ARRANGEMENTS_CONFIRMED|CLOSED|CANCELLED; createdAt; updatedAt; version:int.

Indexes / constraints: weddingId+status+createdAt.

additionalGuestRequestEvents — RSVP

Fields: weddingId:ObjectId; requestId:ObjectId; action; actorId; detailsSnapshot; occurredAt.

Indexes / constraints: Append-only.

weddingBudgets — Finance

Fields: weddingId:ObjectId unique; totalBudgetPaise:int64; reservePaise:int64; currency:'INR'; version:int; updatedAt; updatedBy:ObjectId.

Indexes / constraints: Reserve is inside total budget; check reserve<=total.

budgetCategories — Finance

Fields: weddingId:ObjectId; name:string; allocatedPaise:int64; committedPaise?:int64; paidPaise?:int64; sortOrder:int; status:ACTIVE|ARCHIVED; version:int.

Indexes / constraints: Unique active normalised category name per wedding; aggregates rebuildable.

budgetGovernanceRequests — Finance

Fields: weddingId:ObjectId; requestType; sourceCategoryId?; targetCategoryId?; amountPaise:int64; proposalSnapshot; approvals; status; createdAt; version:int.

Indexes / constraints: Dual-owner approvals for allocation changes and threshold-governed reserve transfers.

budgetReserveAllocations — Finance

Fields: weddingId:ObjectId; fromCategoryId?; toCategoryId:ObjectId?; amountPaise:int64; requestId:ObjectId; createdBy; createdAt.

Indexes / constraints: Append-only transfer ledger.

expenseEstimates — Finance

Fields: weddingId:ObjectId; categoryId:ObjectId; vendorId:ObjectId?; lowPaise:int64; expectedPaise:int64; highPaise:int64; assumptions; createdBy; timestamps; version:int.

Indexes / constraints: Check low<=expected<=high.

vendorQuotes — Finance

Fields: weddingId:ObjectId; vendorId:ObjectId; quoteAmountPaise:int64?; quoteLines:bounded array?; inclusions; exclusions; taxDetails; terms; validUntil:Date?; receivedAt; status:CURRENT|SUPERSEDED|WITHDRAWN.

Indexes / constraints: Separate from estimate and booking; used by finance/vendor workflows.

expenses — Finance

Fields: weddingId:ObjectId; categoryId:ObjectId; vendorId:ObjectId?; title; beneficiaryMembershipId/userId; createdBy; amountPaise:int64; currency:'INR'; status enum; approvedBy?; approvedAt?; version:int; timestamps.

Indexes / constraints: weddingId+status+createdAt desc+_id desc; creator cannot self-approve.

expenseApprovalEvents — Finance

Fields: weddingId:ObjectId; expenseId:ObjectId; action; actorId; reason; approvalRoute; occurredAt.

Indexes / constraints: Append-only.

expensePayments — Finance

Fields: weddingId:ObjectId; expenseId:ObjectId; amountPaise:int64; currency:'INR'; paidAt:Date; recordedBy; method; reference?; evidenceObjectKey?; createdAt.

Indexes / constraints: weddingId+expenseId+paidAt desc; never silently edit/delete.

expenseAdjustments — Finance

Fields: weddingId:ObjectId; expenseId:ObjectId; adjustmentType; amountPaise:int64; reason; requestedBy; approvals; status; createdAt; appliedAt?

Indexes / constraints: Explicit correction/reversal; approval required by policy.

financeAuditEvents — Finance

Fields: weddingId:ObjectId; entityType; entityId:ObjectId; action; actorId; beforeSummary?; afterSummary?; reason?; occurredAt.

Indexes / constraints: Append-only; finance access restricted.

vendorReferences — Vendors

Fields: provider:GOOGLE_PLACES|MANUAL; providerPlaceId:string?; displayName; category; locationSummary; sourceMetadata; timestamps.

Indexes / constraints: Provider reference/catalogue data; not a wedding booking.

weddingVendors — Vendors

Fields: weddingId:ObjectId; vendorReferenceId:ObjectId?; displayNameSnapshot; categoryId:ObjectId; source:PROVIDER|MANUAL; status:ACTIVE|ARCHIVED; createdBy; timestamps; version:int.

Indexes / constraints: weddingId+categoryId+status.

vendorCategories — Vendors

Fields: weddingId:ObjectId; name; allowsMultipleSelections:boolean; createdAt; updatedAt.

Indexes / constraints: Unique normalised category name per wedding.

vendorServices — Vendors

Fields: weddingId:ObjectId; vendorId:ObjectId; serviceName; functionId:ObjectId?; status:ACTIVE|ARCHIVED; createdAt; updatedAt.

Indexes / constraints: weddingId+vendorId+status.

vendorEnquiries — Vendors

Fields: weddingId:ObjectId; vendorId:ObjectId; status:NOT_CONTACTED|CONTACTED|RESPONSE_RECEIVED|NEGOTIATING|SELECTED|REJECTED; managedBy; notes; timestamps; version:int.

Indexes / constraints: weddingId+vendorId+status.

vendorEstimates — Vendors

Fields: weddingId:ObjectId; vendorId:ObjectId; low/expected/high Paise; assumptions; createdAt; updatedAt.

Indexes / constraints: Validate ordered estimate range.

vendorAvailabilityChecks — Vendors

Fields: weddingId:ObjectId; vendorId:ObjectId; functionId/date; state:NOT_CHECKED|AVAILABLE_CONFIRMED|UNAVAILABLE_CONFIRMED|TENTATIVE|UNKNOWN; checkedBy; checkedAt; notes.

Indexes / constraints: Never infer availability from missing record.

vendorSelections — Vendors

Fields: weddingId:ObjectId; categoryId:ObjectId; vendorId:ObjectId; selectedBy; selectedAt; status:SELECTED|REMOVED; version:int.

Indexes / constraints: Partial unique active category selection unless category allows multiple.

vendorBookingRecords — Vendors

Fields: weddingId:ObjectId; vendorId:ObjectId; bookingState:NOT_BOOKED|BOOKING_PENDING|BOOKED|CONTRACTED|CANCELLED; contractReference?; bookingDate?; updatedBy; version:int.

Indexes / constraints: Selection/quote does not imply booked.

vendorActivityEvents — Vendors

Fields: weddingId:ObjectId; vendorId:ObjectId; action; actorId; detailsSnapshot; occurredAt.

Indexes / constraints: Append-only.

shagunSettings — Shagun

Fields: weddingId:ObjectId unique; providerId; destinationReferenceEncrypted; destinationDisplayMasked; status:INACTIVE|PENDING_APPROVAL|ACTIVE|SUSPENDED; version:int; updatedAt.

Indexes / constraints: Exactly one destination per wedding.

shagunConfigurationRequests — Shagun

Fields: weddingId:ObjectId; proposedDestinationEncrypted; proposedMaskedValue; approvals:[two co-owners]; status; createdBy; createdAt.

Indexes / constraints: Dual approval for activation and every destination change.

shagunPaymentIntents — Shagun

Fields: weddingId:ObjectId; guestId:ObjectId; providerId; providerIntentId; amountPaise:int64; currency:'INR'; status; idempotencyKey; timestamps; version:int.

Indexes / constraints: Unique providerId+providerIntentId; provider amount limits apply.

shagunContributions — Shagun

Fields: weddingId:ObjectId; guestId:ObjectId; paymentIntentId:ObjectId; providerPaymentId; amountPaise:int64; currency:'INR'; verifiedAt; status:VERIFIED|REVERSED|DISPUTED; createdAt.

Indexes / constraints: Only provider-verified events create verified contribution.

shagunProviderEvents — Shagun

Fields: providerId; providerEventId; weddingId:ObjectId; payloadHash; receivedAt; processingStatus; processedAt?; payloadRedacted?

Indexes / constraints: Unique providerId+providerEventId; do not store sensitive credentials.

shagunReconciliationEvents — Shagun

Fields: weddingId:ObjectId; contributionId/paymentIntentId; action; actorId; providerStatus; notes; occurredAt.

Indexes / constraints: Append-only; manual note/check does not mark success.

shagunAcknowledgements — Shagun

Fields: weddingId:ObjectId; contributionId:ObjectId; guestId:ObjectId; issuedAt; acknowledgementType.

Indexes / constraints: Unique contribution+acknowledgementType.

shagunAuditEvents — Shagun

Fields: weddingId:ObjectId; action; actorId; entityId:ObjectId; detailsSummary; occurredAt.

Indexes / constraints: Append-only.

photoAlbums — Photos

Fields: weddingId:ObjectId; functionId:ObjectId; title; createdAt; version:int.

Indexes / constraints: Unique weddingId+functionId; one auto-created album per function.

weddingPhotos — Photos

Fields: weddingId:ObjectId; albumId:ObjectId; functionId:ObjectId; uploaderType; uploaderId; objectKey; derivativeKeys; mimeType; byteSize:int; width:int; height:int; checksum; moderationStatus; exifStripped:boolean; timestamps; version:int.

Indexes / constraints: Private R2 keys only; never store public permanent URL.

photoReports — Photos

Fields: weddingId:ObjectId; photoId:ObjectId; reportedBy; reason; status:OPEN|REVIEWED|DISMISSED; createdAt; reviewedAt?

Indexes / constraints: weddingId+photoId+status.

photoRemovalRequests — Photos

Fields: weddingId:ObjectId; photoId:ObjectId; requestedBy; reason; status:PENDING|APPROVED|REJECTED; reviewedBy?; timestamps.

Indexes / constraints: Co-owners decide guest removal requests.

photoAuditEvents — Photos

Fields: weddingId:ObjectId; photoId:ObjectId; actorId; action; reason?; occurredAt.

Indexes / constraints: Append-only.

notifications — Notifications

Fields: weddingId:ObjectId; recipientMembershipId/userId; category; severity:CRITICAL|HIGH|MEDIUM|INFORMATIONAL; title; bodySummary; deepLink; readAt?; createdAt; dedupeKey.

Indexes / constraints: Index recipient+readAt+createdAt; re-authorise deep links.

notificationPreferences — Notifications

Fields: weddingId:ObjectId; recipientMembershipId:ObjectId; categoryPreferences; updatedAt; version:int.

Indexes / constraints: Unique weddingId+recipientMembershipId.

dashboardSnapshots (optional) — Notifications

Fields: weddingId:ObjectId; dashboardType; metrics; generatedAt; sourceWatermark; expiresAt?

Indexes / constraints: Rebuildable cache only; never authoritative finance source.

auditEvents — Audit/lifecycle

Fields: weddingId:ObjectId; actorId; actorType; action; entityType; entityId; requestId; reason?; beforeSummary?; afterSummary?; occurredAt.

Indexes / constraints: Append-only application policy; record sensitive reads/exports.

securityIncidents — Audit/lifecycle

Fields: weddingId:ObjectId; severity:SEV_1..SEV_4; status:DETECTED|TRIAGED|CONTAINMENT|INVESTIGATION|RESOLUTION|CLOSED; summary; openedBy; assignedTo?; timestamps; version:int.

Indexes / constraints: weddingId+status+severity.

dataAccessRequests — Audit/lifecycle

Fields: weddingId:ObjectId; requesterId; requestType; status; reviewedBy?; createdAt; resolvedAt?

Indexes / constraints: Restricted; audited.

dataCorrectionRequests — Audit/lifecycle

Fields: weddingId:ObjectId; requesterId; targetEntityType; targetEntityId; proposedChanges; status; reviewedBy?; timestamps.

Indexes / constraints: Restricted; reviewed workflow.

weddingLifecycleEvents — Audit/lifecycle

Fields: weddingId:ObjectId; eventType; actorId/system; eventSnapshot; occurredAt.

Indexes / constraints: Append-only.

weddingDeletions — Audit/lifecycle

Fields: weddingId:ObjectId unique; status:QUEUED|IN_PROGRESS|RETRY_PENDING|BLOCKED|COMPLETED|FAILED_REQUIRES_REVIEW; stepStates; requestedAt; dueAt; completedAt?; retryCount; version:int.

Indexes / constraints: Idempotent step tracking; no TTL.

weddingDeletionTombstones — Audit/lifecycle

Fields: weddingId:ObjectId unique; deletedAt; deletionJobId; restoreProtectionState.

Indexes / constraints: Retain outside ordinary tenant deletion sweep; replay after backup restore.

weddingExportRequests — Audit/lifecycle

Fields: weddingId:ObjectId; requestedBy; status:PENDING|GENERATING|READY|EXPIRED|FAILED; objectKey?; expiresAt?; createdAt.

Indexes / constraints: Export access restricted; short-lived download URLs.

cancellationNoticeRecords — Audit/lifecycle

Fields: weddingId:ObjectId; noticeStartedAt; noticeEndsAt; status:ACTIVE|COMPLETED; createdAt.

Indexes / constraints: One active cancellation notice per workspace.

supportRecoveryCases — Platform

Fields: weddingId:ObjectId?; requesterIdentity; caseType; evidenceReferences; status; assignedSupportActor; approvalTrail; timestamps.

Indexes / constraints: Platform-level restricted access; all access audited.

outboxEvents — Infrastructure

Fields: eventId unique; weddingId?; aggregateType; aggregateId; eventType; payloadVersion; payload; status:PENDING|PROCESSING|PUBLISHED|FAILED; occurredAt; publishedAt?; attemptCount; nextAttemptAt.

Indexes / constraints: Created in same transaction as business mutation.

inboxEvents — Infrastructure

Fields: consumerName; source; externalEventId; payloadHash; receivedAt; processedAt?; status.

Indexes / constraints: Unique consumerName+externalEventId.

idempotencyRecords — Infrastructure

Fields: scope; actorId; idempotencyKey; requestHash; status:IN_PROGRESS|COMPLETED|FAILED; responseStatus?; responseReference?; createdAt; expiresAt?

Indexes / constraints: Unique scope+actorId+idempotencyKey; financial retention not short TTL.

migrationHistory — Infrastructure

Fields: migrationId unique; checksum; status:STARTED|COMPLETED|FAILED; startedAt; completedAt?; executedBy; errorSummary?

Indexes / constraints: Run by controlled migration job, not every API instance.

5. Cross-collection constraints

• users: unique emailNormalized.

• weddingMemberships: partial unique userId for ACTIVE or SUSPENDED memberships; validate MongoDB partialFilterExpression support and index deployment behaviour.

• rsvps: unique (weddingId, functionId, guestId).

• invitations: unique tokenHash.

• functionPublicationSnapshots: unique (weddingId, functionId, snapshotVersion).

• photoAlbums: unique (weddingId, functionId).

• shagunSettings: unique weddingId.

• shagunProviderEvents: unique (providerId, providerEventId).

• outboxEvents: unique stable eventId.

• idempotencyRecords: unique (scope, actorId, idempotencyKey).

• Compound indexes should support actual filters/sorts and normally start with weddingId. Do not add a redundant single-field weddingId index when a compound index already serves the same access pattern.

• Use explain executionStats on high-value queries; index design must be query-driven.

6. TTL and retention

• TTL only for email verification, password reset, persisted OTP challenge and safe temporary upload reservations.

• No TTL for invitations, audit, payment, RSVP history, expense history or deletion records.

• TTL is asynchronous; enforce expiry immediately in application logic.

• Workspace deletion is a dedicated idempotent workflow; document backup/legal/provider exceptions.

7. Transaction boundaries

• Workspace creation + first membership + audit/outbox; invitation acceptance + membership transition + audit/outbox.

• Governance approvals, function publication, RSVP mutation, expense approval and payment webhook processing use transactions for their atomic invariants.

• Photo moderation uses conditional state transition + audit/outbox. Workspace deletion uses retryable idempotent steps.

• Never call external providers inside a MongoDB transaction.

8. Production implementation gates

• Implement Mongoose schemas and MongoDB JSON Schema validators in the owning NestJS modules.

• Run controlled migrations; use expand-and-contract for breaking changes.

• Test tenant isolation, unique indexes, transactions, OCC, webhook idempotency, deletion and restore tombstones.

• Verify Atlas capabilities, legal retention, key management, UPI provider contracts and R2 residency before production.